

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0748

Newmark 138 kV
March 31, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0748. The Interconnection Customer proposes a 91.7 MW solar photovoltaic facility interconnecting at the Newmark 138 kV substation in the Delaware area. The request was received on February 21, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 337 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,162,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
VIENNA OPERATIONS 345/138 kV Transformer	VIENNA OPERATIONS SUBSTATION / EASTON 2 345 kV Line	131.3	552	9.8
SADDLEBROOKE RANCH / SUNSET 138 kV	EASTON 2 / VIENNA OPERATIONS 345 kV	128.9	512	9.4
CADY / EASTON 2 345 kV	HOLSTON DAM / NAVY MOLE 138 kV	122.5	1,185	13.9
BRISTOL / LIMON 138 kV	CADY / EASTON 2 345 kV	119.6	258	14.9
NINEMILE PT / N LEBANON 138 kV	NINEMILE PT / N LEBANON 138 kV Ckt 1	110.4	288	11.4
SENECA / BANTA CABONA 138 kV	DODD RD / CADY 345 kV line	109.6	277	16.2
NINEMILE PT / N LEBANON 138 kV	N LEBANON / NINEMILE POINT 138 kV Line 2	108.0	294	11.4
LAS LOMAS / NEWMARK 138 kV	VIENNA OPERATIONS / EASTON 2 345 kV	107.8	209	39.1
RIPPLE 345/138kV XFMR	SOUTHTOWN SUBSTATION / VIENNA OPERATIONS SUBSTATION 345 kV L	106.7	433	15.5
WHITE BLUFF / ORANGE BULK 138 kV L	EASTON 2 SUBSTATION / CADY SUBSTATION 345 kV line	105.4	242	6.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VIENNA OPERATIONS 345/1kV Transformer	Replace transformer with higher-rated unit	\$13,815,000	1% (\$113,000)
SUNSET / SADDLEBROOKE RANCH 138 kV	Rebuild line with higher-capacity conductor	\$8,115,000	1% (\$60,000)
CADY / EASTON 2 345 kV	Rebuild line with higher-capacity conductor	\$119,970,000	1% (\$871,000)
BRISTOL / LIMON 138 kV	Rebuild line with higher-capacity conductor	\$4,839,000	4% (\$191,000)
NINEMILE PT / N LEBANON 138 kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,646,000	4% (\$66,000)
SENECA / BANTA CARBONA 138 kV	Reconductor line	\$1,465,000	3% (\$50,000)
NINEMILE PT / N LEBANON 138 kV	Reconductor line	\$439,000	4% (\$18,000)
LAS LOMAS / NEWMARK 138 kV	Reconductor line	\$4,372,000	11% (\$494,000)
RIPPLE 345/138kV XFMR	Replace transformer with higher-rated unit	\$11,792,000	2% (\$200,000)
WHITE BLUFF / ORANGE BULK 138 kV L	Reconductor line	\$4,859,000	2% (\$99,000)
	Total allocated to project		\$2,162,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.